Unit 12: The Laplace Transform

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Condensed Cheat Sheet: Transforms, Inversion, Discontinuous Forcing, and the s-Plane

Unit 12 at a Glance: Core Sub-Topics

The Laplace transform turns a differential equation in t into an algebraic equation in s: initial data is baked in, discontinuous and impulsive forcing become algebra, and a system's behavior is read off the location of its poles. The whole unit is one pipeline — transform, solve the algebra, invert.

- **Definition and existence** — the integral $F(s) = \int_0^\infty e^{-st} f\,dt$ and exponential order.
- **The toolkit** — linearity, the two shift theorems, derivatives, and integrals.
- **Inversion** — partial fractions and completing the square.
- **Discontinuous forcing** — Heaviside steps and Dirac impulses.
- **Convolution and periodicity** — products in s, repeating inputs.
- **The s-plane** — the Fourier connection, poles, and stability.

Part I — Condensed Cheat Sheet

12.1 Intuition and Motivation

The transform re-expresses a signal $f(t)$ in the **frequency domain** $F(s)$: each value of s measures "how much of the mode e^{st} lives inside f." Oscillations and decay rates in t become the locations of poles in s.

Why Bother

The payoff is that differentiation becomes multiplication by s. An ODE (hard calculus) turns into a polynomial equation (easy algebra), which is why the transform solves initial-value problems mechanically.

12.2 Definition and Computing Transforms

The Defining Integral

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt.$$

Base cases straight from the integral:

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}.$$

The trig pair follows from the Euler trick ($a = i\omega$ in $1/(s-a)$):

$$\mathcal{L}\{\cos\omega t\} = \frac{s}{s^2+\omega^2}, \qquad \mathcal{L}\{\sin\omega t\} = \frac{\omega}{s^2+\omega^2}.$$

12.3 Properties, Linearity, Existence, and Inverses

Linearity and Existence

$$\mathcal{L}\{af+bg\} = aF + bG, \qquad \mathcal{L}^{-1}\{aF+bG\} = af + bg.$$

$F(s)$ exists when f is piecewise continuous and of **exponential order** ($|f| \le Me^{at}$); then $F(s) \to 0$ as $s \to \infty$.

Common Pitfall: A Transform Must Vanish at Infinity

If $\lim_{s\to\infty} F(s) \neq 0$, then F is not the transform of a well-behaved f. A function like e^{t^2} grows faster than every Me^{at} and has no transform.

12.4 Transforms of Derivatives and Integrals

Derivative and Integral Rules

$$\mathcal{L}\{f'\} = sF - f(0), \qquad \mathcal{L}\{f''\} = s^2F - sf(0) - f'(0),$$
$$\mathcal{L}\left\{\int_0^t f\right\} = \frac{F(s)}{s}, \qquad \mathcal{L}\{t^n f\} = (-1)^n F^{(n)}(s).$$

Each derivative adds one power of s and peels off one initial value — this is the rule that pulls the initial conditions into the algebra immediately.

12.5 Inverse Transforms and Partial Fractions

Partial-Fraction Playbook

1. **Distinct real root** $(s-r)$: $\dfrac{A}{s-r} \to Ae^{rt}$.
2. **Repeated root** $(s-r)^m$: $\dfrac{A_k}{(s-r)^k} \to A_k\dfrac{t^{k-1}}{(k-1)!}e^{rt}$.
3. **Irreducible quadratic**: complete the square to $(s+a)^2+\omega^2$, then match shifted $\sin / \cos$.

Common Pitfall: Complete the Square, Don't Go Complex

For a quadratic with complex roots, write $s^2+2s+5 = (s+1)^2+4$ and invert to e^{-t} times trig in $2t$. Factoring into complex linear terms is correct but error-prone.

12.6 Solving Initial Value Problems

The Five-Step Method

1. **Transform** the ODE (derivative rules pull in the data).
2. **Substitute** the given $y(0), y'(0)$.
3. **Solve** algebraically for $Y(s)$.
4. **Decompose** by partial fractions / completing the square.
5. **Invert** term by term.

12.7 Heaviside and Piecewise Functions

Unit Step and the Second Shift

$$u(t-c) = \begin{cases} 0, & t < c \\ 1, & t \geq c \end{cases}, \qquad \mathcal{L}\{u(t-c)\} = \frac{e^{-cs}}{s},$$
$$\mathcal{L}\{u(t-c)f(t-c)\} = e^{-cs}F(s).$$

Common Pitfall: Shift the Argument Too

$\mathcal{L}\{u(t-c)f(t)\} \neq e^{-cs}F(s)$. Re-express every t as $(t-c)+c$ first. For t^2 switched on at $c=3$: $t^2 = (t-3)^2 + 6(t-3) + 9$, then transform each piece.

12.8 The Dirac Delta and Impulse Forcing

Sifting and the Impulse Transform

$$\int_{-\infty}^{\infty} \delta(t-c)g(t)\,dt = g(c), \qquad \mathcal{L}\{\delta(t-c)\} = e^{-cs}, \quad \mathcal{L}\{\delta(t)\} = 1.$$

The delta is the derivative of the step: $\delta(t-c) = \frac{d}{dt}u(t-c)$.

Impulse Response

A δ-kick produces a jump in the highest-order state: for $y'' + \cdots = \delta(t-c)$ the velocity y' jumps by 1 at $t=c$ while y stays continuous. The resulting motion is the system's impulse response.

12.9 Convolution and Periodic Functions

Convolution and Periodicity

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau, \qquad \mathcal{L}\{f * g\} = F(s)G(s).$$

For a T-periodic f: $\mathcal{L}\{f\} = \dfrac{1}{1-e^{-sT}} \displaystyle\int_0^T e^{-st} f(t)\,dt.$

Common Pitfall: Products Don't Multiply

$\mathcal{L}\{f\,g\} \neq F(s)G(s)$. The product $F(s)G(s)$ inverts to the *convolution* $f * g$, never to the pointwise product $f(t)g(t)$.

12.10 The Fourier Connection

Writing $s = \sigma + i\omega$, the transform is a Fourier transform of the damped signal $e^{-\sigma t} f(t)$:

$$F(s) = \int_0^\infty \left[e^{-\sigma t} f(t)\right] e^{-i\omega t}\,dt.$$

The Imaginary Axis Is the Fourier Axis

Evaluating $H(s)$ on $s = i\omega$ gives the frequency response $H(i\omega)$. The real part σ is the convergence knob Fourier lacks — it lets Laplace handle growing and switched-on transients.

12.11 Frequency Domain and Pole Diagrams

Poles and Behavior

A pole at $s = \alpha + i\beta$ contributes a mode $e^{\alpha t}\cos(\beta t)$:

$$\alpha < 0 \Rightarrow \text{decay (stable)}, \quad \alpha = 0 \Rightarrow \text{oscillation (marginal)}, \quad \alpha > 0 \Rightarrow \text{growth (unstable)}.$$

Common Pitfall: Resonance Is a Repeated Pole

A simple pole on the imaginary axis gives bounded oscillation; a **double** pole there is what produces t-growth. Check pole *multiplicity*, not just location.

12.12 Deriving the Transform from First Principles

The transform is the continuous limit of a power series. Replacing the discrete $\sum a_n x^n$ with the integral $\int_0^\infty f(t)\,x^t\,dt$ and writing $x = e^{-s}$ (so $x^t = e^{-st}$) yields exactly $\int_0^\infty e^{-st} f(t)\,dt$.

Unit Key Takeaway

The Laplace transform is one disciplined pipeline: transform $\rightarrow$ solve the algebra $\rightarrow$ invert. Initial data rides in through the derivative rules, discontinuities become e^{-cs} and δ bookkeeping, products become convolutions, and the geometry of the poles tells you whether a system decays, oscillates, or blows up before you invert a single term.